

Martin A. Fernandez

mafern@colostate.edu – <https://mafern.github.io/>

EMPLOYMENT

Current – Research Scientist – Computing and Data Sciences, Boston University
2024 - 2025 – Research Scientist – Atmospheric Science, Colorado State University
2023 - 2024 – Postdoctoral Fellow – Atmospheric Science, Colorado State University

EDUCATION

2023 PhD in Physics – University of California Riverside
2018 M.S. in Physics – University of California Riverside
2017 B.S. in Physics – Western Washington University

CURRENT RESEARCH

Using interpretable and/or explainable (posthoc XAI) machine learning methods to improve climate and weather prediction:

- Probabilistic machine learning methods for forecasting tropical cyclone track and intensity error. *With Dr. Elizabeth Barnes and Dr. Mark DeMaria.*
- Multi-year to decadal climate prediction using machine-learning assisted model-analogs. *With Dr. Elizabeth Barnes.*
- Novel methods to separate the forced response from internal variability in observations. *ForceSMIP collaboration.*

RECENT PRESENTATIONS & WORKSHOPS

AMS Annual Meeting Jan 2025 (New Orleans)
Multi-Year-to-Decadal Temperature Prediction using Machine Learning Model-Analogs
Earth System Predictability Across Timescales April 2024 (NCAR)
Multi-Year to Decadal Analog Forecasting using Machine Learning Derived Masks
Confronting Earth System Models with Observations March 2024 (NCAR)
Ranking Earth System Models using Observations and Machine Learning Analog Forecasting
AMS Annual Meeting Jan/Feb 2024 (Baltimore)
Predicting Tropical Cyclone Track Forecast Errors Using a Probabilistic Neural Network
ForceSMIP Hackathon Aug 2023 (NCAR)
Developing Methods for Separating the Forced Response from Internal Variability

OUTREACH AND MENTORSHIP

Primary mentor for REU student 2024
Founder/organizer for CSU/CIRA Research Staff Events 2023 – 2025
Co-mentor for REU student 2023
Co-mentor for undergraduate summer research project 2023

Advisor/mentor for high school student	2018 – 2020
Co-founder of UCR Physics Organization for Women and the UnderRepresented	2018 – 2020
Host of public night sky observing	2015 – 2017
WWU Women in Physics	2015 – 2017

PROGRAMMING SKILLS

expert: Python, L^AT_EX, and High-performance computing.

familiar: C, C++, Mathematica, IDL, and HTML/CSS.

AWARDS

NSF GRFP (Graduate Research Fellowship Program)	2019 – 2023
UCR Chancellor’s Distinguished Fellowship	2017
WWU Alumni Association Leader Scholarship	2016

PAST RESEARCH

2018 – 2023: Using cosmological simulations and machine learning to explore beyond-standard-model physics and constrain cosmological & astrophysical parameters. *With Dr. Simeon Bird (University of California Riverside).*

2016 – 2017: Theory & modeling of guided wave plasmon polariton modes on novel waveguide architectures, with applications to solar cells and medical devices. *With Dr. Brad Johnson (Western Washington University).*

2015 – 2017: Identifying & characterizing pre-main sequence double-lined spectroscopic binaries in young star forming environments. *With Dr. Kevin Covey (Western Washington University).*

FIRST AUTHOR PUBLICATIONS

- Fernandez, M. A. and Barnes, E. A., *Multi-Year-to-Decadal Temperature Prediction using a Machine Learning Model-Analog Framework*, **submitted to Machine Learning: Earth**, [arXiv: 2502.17583](#).
- Fernandez, M. A., Barnes, E. A., Barnes, R. J., DeMaria, M., McGraw, M., Chirokova, G., and Lu, L., *Predicting Tropical Cyclone Track Forecast Errors using a Probabilistic Neural Network*, **AIES**. [arXiv: 2503.09840](#).
- Fernandez, M. A., Bird, S., and Ho, M.-F., *Cosmological Constraints from the eBOSS Lyman- α Forest using the PRIYA Simulations*, **JCAP**. [arXiv: 2309.03943](#).
- Fernandez, M. A., Ho M.-F., and Bird, S., *A Multi-fidelity Emulator for the Lyman- α Forest Flux Power Spectrum*, **MNRAS**. [arXiv: 2207.06445](#).
- Fernandez, M. A., Bird, S., and Upton Sanderbeck, P., *Effect of separate initial conditions on the Lyman- α forest in simulations*, **MNRAS**. [arXiv: 2009.09119](#).
- Fernandez, M. A., Bird, S., and Cui, Y., *Cosmic Filaments from Cosmic Strings*, **Phys. Rev. D**. [arXiv: 2004.13752](#).
- Fernandez, M. A., Covey, K. R., De Lee, N., et al., *Identification and Radial Velocity Extraction for 100+ Double-Lined Spectroscopic Binaries in the APOGEE/IN-SYNC Fields*, **PASP**. [arXiv: 1706.01161](#).

CO-AUTHOR PUBLICATIONS

- Wills, R. J., Deser, C., McKinnon, K., Phillips, A., Po-Chedley, S., ..., Fernandez, M., et al., *A community estimate of the multi-variate forced climate response from observations*, **in preparation**.
- Keys, P., Fernandez, M. A., DesRosiers, A., Connolly, C., Hueholt, D., Hughes, A.-C, et al., *Decarbonization In SCenariOs (DISCO)*, **in preparation**.
- Ho, M.-F, Qezlou, M., Bird, S., Yang, Y., Avestruz, C., Fernandez, M. A., and Irsic, V., *Small-scale Lyman alpha forest cosmology with PRIYA: Constraints from XQ100 and KODIAQ-SQUAD one-dimensional flux power spectra*, [arXiv: 2509.18271](#).
- Ennis, K. E., Barnes, E. A., Arcoda, M. C., Fernandez, M. A., Maloney, E. D., *Turning Up the Heat: Assessing 2-m Temperature Forecast Errors in AI Weather Prediction Models During Heat Waves*, **submitted to Royal Meteorological Society: Meteorological Applications**, [arXiv: 2504.21195](#).
- Wills, R. J., Deser, C., McKinnon, K., Phillips, A., Po-Chedley, S., ..., Fernandez, M., et al., *Forced Component Estimation Statistical Method Intercomparison Project (ForceSMIP)*, **submitted to Journal of Climate**, [ESS Open Archive](#).
- Rader, J. K., Connolly, C., Fernandez, M. A., and Gordon, E. M., *Attribution of the record-high 2023 SST using a deep-learning framework*, **Environmental Research Communications**.
- Tohfa, H., Bird, S., Ho, M.-F, Qezlou, M., and Fernandez, M., *Forecast Cosmological Constraints with the 1D Wavelet Scattering Transform and the Lyman- α forest*, **Phys. Rev. Letters**, [arXiv: 2310.06010](#).
- Bird, S., Fernandez, M., Ho, M.-F, Qezlou, M., Monadi, R., et al., *PRIYA: A New Suite of Lyman- α Forest Simulations for Cosmology*, **JCAP**, [arXiv: 2306.05471](#).
- Ho, M.-F, Bird, S., Fernandez, M., and Shelton, C. R., *MF-Box: Multi-fidelity and multi-scale emulation for the matter power spectrum*, **MNRAS**, [arXiv: 2306.03144](#).
- Troup, N. W., Nidever, D. L., De Lee, N., Carlberg, J., Majewski S. R., Fernandez, M., et al., *Companions to APOGEE Stars. I. A Milky Way-spanning Catalog of Stellar and Substellar Companion Candidates and Their Diverse Hosts*, **AJ**, [arXiv: 1601.00688](#).